cre8: whole-stack AWS cost model

Prepared: 11 November 2026
Status: planning estimate, not a deployment approval or capacity guarantee
Decision unchanged: stay on Render now; AWS remains later scope.

UK planning context: see the UK business plan (business-plan-uk.pdf) and licensing/payment checklist (uk-licensing-and-payments.pdf). At the explicit planning assumption of USD 1 = GBP 0.80, the USD 600/825 budgets translate to GBP 480/660. This is not a live exchange rate or a London/Ireland-region quote. UK consumer VAT, cloud invoice taxes and music licensing need separate treatment.

Executive summary

At the production spike's proposed launch envelope, allow **US$600/month for a lean AWS stack**, including a small staging environment and approximately 20% contingency. A future two-AZ application/database topology needs approximately **US$825/month** on the assumptions below. Neither figure includes tax, paid support, engineering time or a warm disaster-recovery region.

Option	Modelled recurring subtotal	Subtotal + 20%	Recommended monthly budget
Lean, one API instance, single-AZ database	$479.96	$575.95	**$600**
Resilient target, two API instances, Multi-AZ database	$671.52	$805.82	**$825**

These are complete planning budgets rather than the earlier conversation's $300-450 application-hosting estimate. They explicitly add private-subnet networking, staging, backup copies, monitoring, build delivery and operational allowances. Some rows are calculated from published rates; others are deliberately labelled allowances pending an AWS Calculator quote and measured workloads.

The resilient option is a **future architecture**, not something the current PostgreSQL compatibility backend can safely enable merely by adding replicas. Paying for redundancy does not establish the spike's availability or recovery objectives.

1. Scope and pricing basis

- Currency: USD, before VAT/sales tax. No exchange-rate assumption.
- Reference region: US East (N. Virginia), `us-east-1`; audience mainly Europe/US. Requote regional services for London/Ireland before choosing a region.
- Billing: 730 hours/month for always-on resources; 30 days for workload maths.
- On-demand Linux/x86 Fargate; no Spot, Savings Plans, reserved instances, promotional credits or new-account free-trial assumptions.
- One production CloudFront distribution and root domain; small staging is not given its own paid CloudFront subscription.
- Frontend, PWA, admin and guides share the web/API delivery infrastructure. There is no separate per-user React hosting or admin licence charge.
- PostgreSQL remains the source of truth for metadata, sessions and durable jobs. No Redis, OpenSearch, Kubernetes, Aurora, RDS Proxy or SQS is assumed.
- Private S3 holds media and static assets; temporary work uses task-local disk. No EFS or permanent media disk is included.
- Public HTTPS ALB in two AZs; private API/worker tasks and a private database. S3 gateway endpoints keep S3 traffic off NAT. No paid interface endpoints are assumed; adding those requires a separate quote.

2. Workload inherited from the spike

These are **unapproved planning/test inputs**, not observed production traffic.

Input	Assumption
Community	1,000 registered users; 100 active sessions/sockets
Catalogue	10,000 files, 1,000 packs, 100,000 messages
Original media	25 MB average per file: approximately 250 GB
Ingest	20 uploads/day at 25 MB: approximately 15 GB/month
Listening	100 simultaneous listeners at 320 kbit/s for 8 hours/day
Metadata traffic	20 requests/second sustained; short bursts to 50/second
Recovery targets	Metadata RPO <=15 minutes; independent-original-backup RPO <=24 hours; RTO <=4 hours

Input	Assumption
Reliability objective	99.9% successful non-user-error requests; not an established SLA

Listening calculation, using decimal network units:

```
100 listeners x 320,000 bits/second / 8 = 4,000,000 bytes/second
4 MB/second x 3,600 = 14.4 GB/hour
14.4 GB/hour x 8 hours/day x 30 days = 3,456 GB/month

20 metadata requests/second x 86,400 x 30 = 51,840,000 requests/month
```

Budget roughly 55-65 million total monthly viewer requests after frontend, audio range requests, artwork and reconnects. That request estimate needs measurement: audio delivery does not necessarily mean one GET per track, and WebSocket frames must not be counted as individual HTTP requests.

Downloads, repeat downloads and ZIP archives are **additional to** the 3.46 TB listening calculation. Previews and downloads must be measured independently.

3. Proposed stack being priced

```
Browser / PWA / admin / guides
  -> CloudFront Business: TLS, CDN, WAF, DNS
     -> private S3: static assets and explicitly authorized media
     -> HTTPS ALB
        -> private Fargate Node API: HTTP + WebSockets
           -> private RDS PostgreSQL
           -> private S3
        -> private Fargate media worker
           -> PostgreSQL durable job leases
           -> private S3

Separate backup account/region:
  immutable/versioned media copies + protected database recovery copies

Delivery/operations:
  ECR + CI builds + Secrets Manager + KMS + CloudWatch + small staging
```

Keep `/api`, session cookies, pack/share URLs and `/admin` stable. Cache public static assets separately from personalized API responses. Private media needs explicit authorization and appropriately scoped signed delivery; an opaque S3 key or a public CDN URL is not an access check.

Compute assumptions

- Lean API: one task, 1 vCPU / 2 GiB, always on.
- Resilient API: two such tasks across two AZs.
- Lean worker: one task, 0.5 vCPU / 1 GiB, always on.
- Resilient worker: one task, 1 vCPU / 2 GiB, restartable with durable jobs. This is not a redundant pair of workers; queued work must survive restart.
- Lean database allowance targets a `db.t4g.medium`-class PostgreSQL instance with 20 GB gp3, single-AZ. Resilient allowance targets a primary/standby Multi-AZ instance deployment, not a three-instance Multi-AZ DB cluster.
- Staging uses synthetic, much smaller fixtures and a smaller separate database. It is not a second production-sized load-test environment.

The API and worker allocations are starting points, not proof of capacity. Five concurrent ffmpeg processes are not assumed safe at these sizes: queue the five-upload burst and enforce measured worker concurrency/RSS/temp limits. RDS burst CPU-credit charges and sustained database pressure require monitoring.

4. Complete recurring monthly estimate

C = calculated from quantities/rates; **A** = planning allowance, not a fixed service quote. AWS regional/SKU prices must be checked before provisioning.

Component	Basis	Lean	Resilient
API Fargate compute	C: 1 or 2 x 1 vCPU / 2 GiB x 730 hours	$36.04	$72.08
Media worker compute	C: 0.5 vCPU / 1 GiB, or 1 vCPU / 2 GiB	$18.02	$36.04
RDS PostgreSQL + 20 GB gp3	A: single-AZ or primary/standby Multi-AZ; verify engine/region quote	$60.00	$120.00
CloudFront Business	C: one plan, 50 TB and 125M requests/month	$200.00	$200.00

Component	Basis	Lean	Resilient
ALB base + LCU usage	C/A: $16.43 base plus metadata/upload processing headroom	$22.00	$25.00
ALB public IPv4	C: two addresses x $0.005/hour x 730	$7.30	$7.30
NAT gateways + their IPv4	C: one AZ or two AZs; $32.85 + $3.65 each	$36.50	$73.00
NAT data + inter-AZ transfer	A: low-volume control traffic; media uses S3 gateway endpoints	$2.00	$4.00
Primary S3 storage	C: 400 GB x $0.023/GB-month	$9.20	$9.20
Independent media backup storage	C: 300 GB x $0.023/GB-month at a similarly priced destination	$6.90	$6.90
S3 requests + incremental backup transfer	A: GET/PUT/LIST, lifecycle/inventory and 15 GB/month cross-region additions	$3.00	$5.00
Extra database backup/copy storage	A: beyond included RDS automated-backup allowance	$5.00	$10.00
KMS keys and requests	A: few customer-managed keys; S3 Bucket Keys where compatible	$3.00	$4.00
Application logs, metrics, alarms, checks	A: bounded logging, retention and querying	$15.00	$22.00
Secrets Manager	A: few secrets, cached retrieval; not per-request fetching	$2.00	$3.00
ECR image retention	A: small image history with lifecycle cleanup	$1.00	$2.00
CI build execution	A: modest build minutes; use existing CI or CodeBuild, not both	$5.00	$8.00
Transactional email reserve	A: optional low-volume SES; not currently a newly implemented feature	$1.00	$2.00
Domain registration amortization	A: renewal spread monthly; actual TLD/registrar varies	$2.00	$2.00
Small staging environment	A: breakdown below; shares production networking capacity, not data	$45.00	$60.00
Total before contingency		**$479.96**	**$671.52**
20% contingency	Covers normal usage/price uncertainty, not unlimited growth	**$95.99**	**$134.30**
Total with contingency		**$575.95**	**$805.82**
Rounded monthly budget		**$600**	**$825**

Fargate rate calculation: approximately $0.0404784/vCPU-hour and $0.004446/GiB-hour from AWS's Linux/x86 Northern Virginia pricing example. One 1-vCPU/2-GiB task costs approximately `730 x (0.0404784 + 2 x 0.004446) = $36.04`. The included default 20 GiB ephemeral storage is sufficient only if admission and cleanup keep scratch use within it; larger allocations add cost.

Staging allowance

The lean $45 consists of approximately $9 for a 0.25-vCPU/0.5-GiB task, $26 for a small separate PostgreSQL database/storage, $3 for logs, $2 for synthetic media/backups, $1 for secrets and $4 for incremental networking/operations. The resilient column adds $15 for more test-runtime headroom.

Use the existing ALB with a separate target group and protected staging hostname, and shared NAT capacity; isolate database, buckets/prefix permissions and secrets. These are separate environments but **not separate VPC/account isolation**. A dedicated staging ALB, NAT, account or CloudFront plan is not included. Protect direct staging ingress; do not make test data/admin endpoints public. Large load tests and temporary restore environments are extra.

Avoid double counting

- CloudFront Business includes WAF capabilities, DNS, edge TLS and certain delivery/WAF log ingestion. Those are not charged a second time here. Additional WAF products, premium rules and application logging can still cost.
- ACM-managed public certificates for the ALB are assumed without a separate certificate fee; paid certificate products are not included.
- Business advertises **1 TB of S3 Standard storage credits**. The table conservatively charges all 700 GB of primary/backup storage anyway. If the full credit applies, it could remove up to **$16.10/month** here. Do not assume a credit in the production account pays the separate backup account's bill; confirm scope and eligibility.
- Transfer from supported AWS origins to CloudFront is not also billed as normal internet egress. S3 GET requests, backup replication and NAT processing remain separate where applicable.
- IAM roles, VPC/security groups, ECS orchestration on Fargate and PostgreSQL software do not add separate licence/subscription lines.

5. Why CloudFront Business is the baseline

Current published flat-rate plans list:

Plan	Monthly fee	Viewer requests	Transfer
Pro	$15	10 million	50 TB
Business	$200	125 million	50 TB

The spike's sustained API load alone is approximately 51.84M requests/month, so Pro is not the whole-application budget. Business also includes custom cache and origin-request policies needed for a controllable API/private-media design. Confirm exact support for signed delivery, cookie/header forwarding, range requests and cache-disabled API behaviors before committing to this plan. Never weaken authorization or cache private responses just to fit a cheaper tier.

Allowances are plan boundaries, not a promise of unlimited capacity: monitor usage and AWS's current plan terms. Do not rely on free-tier eligibility. If the required configuration is incompatible with a flat-rate plan, price pay-as-you-go CDN/WAF/DNS/logging together; **this model must then be revised**. No new pay-as-you-go transfer quote is silently substituted here.

6. Storage, backups and growth

The primary 400 GB allowance comprises 250 GB originals plus approximately 150 GB for previews, artwork, static assets, retained versions and short-lived archives. The independent backup allowance is 250 GB originals plus 50 GB for retained changes. These are storage budgets, not measured encoding ratios.

- Use lifecycle rules and bounded ZIP retention; do not retain every archive indefinitely or copy regenerable archives into disaster backups.
- Originals add approximately 15 GB/month. At the reference storage rate, a primary and independent copy add roughly **$0.69/month of recurring storage for each month's new uploads**, before previews, versions and transfer.
- At 12 months of that ingest, originals alone add 180 GB per copy. Primary storage and backup allowances must grow or be repriced; the opening-month table is not a fixed twelve-month storage bill.
- An independent, separately administered backup account/region with restricted deletion and tested retention is assumed. S3 versioning in the primary account alone is not that backup.
- Daily incremental media copies target the accepted <=24-hour original RPO. RDS PITR supports metadata recovery, but remote/cross-account snapshot copies do not automatically provide a <=15-minute regional-disaster RPO.
- Achieving a <=15-minute RPO after loss of the primary region may require continuous cross-region replication and additional database/transfer spend, **not included**. Recovery objectives must state which failure domains apply.
- Retain 30 days of operational backups initially and run restore drills. Retention churn, lock duration, full-copy behavior and restored resources can exceed these small allowances; measure them before making recovery promises.

7. Sensitivity and budget risks

Change	Effect
Listening is only 10 concurrent users for 8 hours/day	About 346 GB/month; infrastructure/Business plan remain mostly fixed at the modelled request load
100 listeners run 24 hours/day at 320 kbit/s	About 10.37 TB/month, before downloads
Streams average 1.411 Mbit/s instead of 320 kbit/s	About 15.24 TB/month at 8 hours/day
Streams average 10 Mbit/s at that utilization	About 108 TB/month; exceeds the assumed 50-TB plan
API averages 50 requests/second all month	About 129.6M API requests before media/static requests; exceeds Business's 125M allowance
Add one 1-vCPU/2-GiB API task	Approximately $36/month compute plus traffic/database effects
Add one NAT gateway with its public IPv4	Approximately $36.50/month before processed data
A small RDS instance cannot sustain the workload	Requote the next size/class; CPU-credit and storage growth charges are additional risks
Each of 51.84M API requests logs 1 KB	Roughly 52 GB of logs before worker/access/error logs; verbose logging can exceed the observability allowance

The current implementation serves original audio to authenticated listeners. Do not silently treat every full stream as a 320-kbit/s preview. Real source bitrates and download behavior are the largest workload uncertainties. Similarly, full community snapshots and collection rewrites can make 20 API requests/second much more expensive than 20 small indexed reads.

8. What is implemented versus still required

Current deployment configuration is a single Render service with a 1-GB disk. The opt-in PostgreSQL compatibility layer preserves existing API contracts, but reads full state, rewrites changed collections, serializes work and keeps WebSocket/push dispatch process-local. Stage A/B proofs are not automatically the production media pipeline.

Before either priced AWS option can be called deployment-ready:

1. Implement and verify private S3 upload/read/delete, quarantine, media validation, preview and archive lifecycles; preserve all access rules.
2. Wire durable processing and commit-safe dispatch into the real API, rather than assuming the Stage B harness is already its production worker.
3. Configure ECS, ALB, task roles, private networking, verified database TLS, secrets, WAF and correctly separated cache behaviors.
4. Load-test actual full-state PostgreSQL/API behavior, source audio bitrates, 100 sockets/listeners and the five-upload burst. Adjust compute/DB budgets.
5. Before multiple APIs: remove unsafe per-process state assumptions, implement cross-instance realtime delivery and durable effects, and verify ordering, idempotency, session behavior and pool limits under failover.
6. Rehearse guarded import, backup/restore, rollback, key recovery and deployment shutdown. Measure the accepted RPO/RTO and define maintenance windows.

The lean option has API/database single points of failure. The resilient option reduces those only after this work; neither topology alone proves 99.9%.

9. Exclusions and spending controls

Not included: migration/build engineering, on-call labour, paid AWS technical support, taxes, payment-provider fees, SMS, marketing email, third-party analytics, large guide-video distribution, dedicated security products, warm DR compute, regional database replica, premium database IOPS, long-term audit retention, unexpected abuse at origins, or duplicate live Render hosting during migration. Small guide/static assets are within the storage/request allowances; large videos must be added to transfer. Web Push uses existing browser push services, not a per-subscriber SNS bill.

Set cost alerts at 50%, 80% and 100% of the chosen budget plus anomaly detection. Billing alerts are **not a hard spending cap**. Enforce upload/download quotas, job concurrency, scratch limits, archive TTLs and log retention in the product. Monitor CloudFront requests/bytes, S3 version growth, NAT bytes, RDS CPU credits, database latency, worker queue age and actual restore costs.

Recommendation: retain Render for the current phase. For AWS approval, carry **$600/month lean** or **$825/month resilient**, then replace allowances with a saved regional AWS Calculator estimate and measured staging results. Provision nothing solely on the strength of this document.

10. Sources and estimate confidence

Public pricing/features checked on 11 November 2026. Dynamic pricing tables, account eligibility and negotiated pricing were not verified in an AWS account.

Source	Used for
Production spike	Workload, Render-first decision, media/recovery requirements
Current Render configuration, container	Current single-service/disk topology and shared frontend/admin hosting
PostgreSQL compatibility storage	Current compatibility limits; not a normalized multi-instance backend
AWS Fargate pricing	CPU/memory calculation and scratch-storage basis
RDS PostgreSQL pricing	Instance/storage/backup/transfer billing; exact SKU allowance still needs a quote
CloudFront plan prices	$15/$200 plans, request and transfer allowances
CloudFront plan features	Custom policies, included services, S3 storage credits and plan restrictions
CloudFront pay-as-you-go	AWS origin-to-CloudFront transfer treatment
ALB pricing	$0.0225/hour base and $0.008/LCU-hour reference example
VPC pricing	NAT $0.045/hour, $0.045/GB reference and public IPv4 charges
S3 pricing	Standard storage/request/replication charging; $0.023/GB-month reference
KMS, Secrets Manager	Key and secret operational allowances
CloudWatch, ECR	Observability and image-retention allowances
SES	Optional low-volume transactional email reserve

Confidence: medium for an opening-month financial envelope; low for capacity until real load tests. Calculated prices are not fixed quotes, and allowances must not be presented as benchmark results.